Again differentiating, we have

$$\frac{s'''}{s'} - \left(\frac{s''}{s'}\right)^2 = S'^2\left[\frac{s_3}{s_1} - \left(\frac{s_2}{s_1}\right)^2\right] + 3S'\frac{s_2}{s_1}\frac{S''}{S'} - \left(\frac{S''}{S'}\right)^2.$$

But

$$-\tfrac{1}{2}\left(\frac{s''}{s'}\right)^2 = S'^2\left[-\tfrac{1}{2}\left(\frac{s_2}{s_1}\right)^2\right] - S'\frac{s_2}{s_1}\frac{S''}{S'} - \tfrac{1}{2}\left(\frac{S''}{S'}\right)^2,$$

and consequently

$$\frac{s'''}{s'} - \tfrac{3}{2}\left(\frac{s''}{s'}\right)^2 = S'^2\left[\frac{s_3}{s_1} - \tfrac{3}{2}\left(\frac{s_2}{s_1}\right)^2\right] + \frac{S'''}{S'} - \tfrac{3}{2}\left(\frac{S''}{S'}\right)^2;$$

that is,

$$\{s,\, x\} = \left(\frac{dS}{dx}\right)^2\{s,\, S\} + \{S,\, x\},$$

the required formula.

In a very similar manner, taking x a function of X, it is shown that

$$\{s,\, x\} = \left(\frac{dX}{dx}\right)^2\{s,\, X\} - \{x,\, X\}.$$

3. If in this formula we write S for s, and substitute the resulting value of $\{S,\, x\}$ in the former formula, we have

$$\{s,\, x\} = \left(\frac{dS}{dx}\right)^2\{s,\, S\} + \left(\frac{dX}{dx}\right)^2\{S,\, X\} - \{x,\, X\},$$

which is the formula for the change of both variables. It in fact includes the other two: viz. writing $X = x$, or $S = x$, and observing that $\{x,\, x\} = 0$, we have the other two formulæ.

4. By putting in the first formula $S = x$, we obtain

$$\{x,\, s\} = -\left(\frac{ds}{dx}\right)^2\{s,\, x\},$$

a formula for the interchange of the variables.

5. Writing $S = \dfrac{as+b}{cs+d}$, and using for a moment the accents to denote differentiation in regard to s, we have

$$S' = \frac{ad-bc}{(cs+d)^2}, \qquad \frac{S''}{S'} = \frac{-2c}{cs+d},$$

and thence

$$\frac{S'''}{S'} - \left(\frac{S''}{S'}\right)^2 = \frac{2c^2}{(cs+d)^2},$$

$$-\tfrac{1}{2}\left(\frac{S''}{S'}\right)^2 = \frac{-2c^2}{(cs+d)^2}.$$

Consequently $\{S,\, s\} = 0$, whence also $\{s,\, S\} = 0$.

Hence in the first formula $\{S,\, x\} = \{s,\, x\}$, that is,

$$\left\{\frac{as+b}{cs+d},\, x\right\} = \{s,\, x\};$$

viz. we may, in the derivative $\{s,\, x\}$, write for s any homographic function $\dfrac{as+b}{cs+d}$ of s.

6. Again, if $X = \dfrac{\alpha x+\beta}{\gamma x+\delta}$, then from the second formula

$$\{s,\, x\} = \frac{(\alpha\delta-\beta\gamma)^2}{(\gamma x+\delta)^4}\{s,\, X\};$$

that is,

$$\left\{s,\, \frac{\alpha x+\beta}{\gamma x+\delta}\right\} = \frac{(\gamma x+\delta)^4}{(\alpha\delta-\beta\gamma)^2}\{s,\, x\};$$

and here, changing s into $(as+b) \div (cs+d)$, we have finally

$$\left\{\frac{as+b}{cs+d},\, \frac{\alpha x+\beta}{\gamma x+\delta}\right\} = \frac{(\gamma x+\delta)^4}{(\alpha\delta-\beta\gamma)^2}\{s,\, x\},$$

which is the formula for the homographic transformation of the two variables s, x.

7. Let s be a given function of x, the equation $\{S,\, x\} = \{s,\, x\}$ is a differential equation of the third order in S, and by what precedes, its general integral is $S = \dfrac{as+b}{cs+d}$.

The direct process is as follows: we have a first integral $\dfrac{S''}{S'} = \dfrac{s''}{s'} - \dfrac{2cs'}{cs+d}$; a second integral $\log S' = \log s' - 2\log(cs+d) + \text{const.}$, that is, $S' = \dfrac{cAs'}{(cs+d)^2}$; and thence a final integral $S = B - \dfrac{A}{cs+d}$, which is equivalent to the foregoing value of S.

The Quadric Function of three or more Inverts. Art. Nos. 8 to 15.

8. We consider a quadric function of any number of inverts $\dfrac{1}{x-a}$, $\dfrac{1}{x-\beta}$, $\ldots$, all of them different: it is assumed that the constant term is $= 0$, and also that the sum of the coefficients of the linear terms is $= 0$. We have therefore square terms $\dfrac{a}{(x-a)^2}, \ldots$, product terms $\dfrac{h}{x-a\,.\,x-\beta}$, and linear terms $\dfrac{A}{x-a}$, where the sum of the coefficients A is $= 0$. Any product term $\dfrac{h}{x-a\,.\,x-\beta}$ is expressible in the form of a difference $\dfrac{h}{a-\beta}\left(\dfrac{1}{x-a} - \dfrac{1}{x-\beta}\right)$ of two linear terms, and (the coefficients of these being equal), after it is thus expressed, the sum of the coefficients of the linear terms is still $= 0$. The function is thus always expressible in the form

$$\frac{a}{(x-a)^2} + \frac{b}{(x-\beta)^2} + \cdots + \frac{A}{x-a} + \frac{B}{x-\beta} + \cdots,$$

where the sum $A + B + \cdots$ is $= 0$: this may be called the reduced form.

9. Observe that any particular invert $\dfrac{1}{x-a}$ may disappear altogether from the reduced form: this will be the case if $a = 0$, that is, if the original form contains no term in $\dfrac{1}{(x-a)^2}$, and if also $A = 0$. An invert thus disappearing from the reduced form is said to be non-essential: and the inverts which do not disappear are said to be essential. The original form contains in appearance the non-essential inverts, but it is really a quadric function of the essential inverts only.

10. Imagine the original function expressed as a rational fraction, the denominator being the product $(x-a)^2(x-\beta)^2(x-\gamma)^2\cdots$ of the squared factors corresponding to all the inverts (non-essential as well as essential): the numerator will be in general of a degree less by 2 than that of the denominator, but the coefficients of any one or more of the higher powers of x may vanish, and the numerator will then be of a lower degree. But this numerator will for any non-essential invert $\dfrac{1}{x-\gamma}$ contain the factor $(x-\gamma)^2$, or, dividing the numerator and denominator each by this factor, the difference of the degrees of the numerator and denominator will remain unaltered; that is, the difference will have the same value whether we do or do not attend to the non-essential inverts; or say it will have the same value for the original form and for the reduced form.

11. It is to be remarked that the linear terms $\dfrac{A}{x-a} + \dfrac{B}{x-\beta} + \dfrac{C}{x-\gamma} + \cdots$, where $A+B+C+\cdots = 0$, can be (and that in a variety of ways) expressed as a sum of differences $\dfrac{1}{x-a} - \dfrac{1}{x-\beta}$, that is, as a sum of product-terms $\dfrac{1}{x-a\,.\,x-\beta}$. Hence the quadric function can be (and that in a variety of ways) expressed as a homogeneous function $(a, b, \ldots \;|\!\cdots\, |\; \frac{1}{x-a}, \frac{1}{x-\beta}, \ldots)^2$; we must have in the form all the essential inverts, and we need have these only. Supposing that this is so, and that the number of the essential inverts is $= n$, then the number of constants is $= \frac{1}{2}n(n+1)$, whereas the number of constants in the reduced form is only $= 2n - 1$: hence the coefficients are not determinate; or, what is the same thing, we may have different quadric functions having each of them the same reduced function; these quadric functions, as having the same reduced function, can only differ by multiples of the evanescent expressions

$$\frac{\beta-\gamma}{x-\beta\,.\,x-\gamma} + \frac{\gamma-a}{x-\gamma\,.\,x-a} + \frac{a-\beta}{x-a\,.\,x-\beta}, \quad \&c.$$

In particular, if the number of essential inverts is $= 3$, then the quadric function is of the form

$$\left(a,\ b,\ c,\ f,\ g,\ h\middle|\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma}\right)^2,$$

which contains one superfluous constant, and equivalent functions differ only by a multiple of

$$\frac{\beta-\gamma}{x-\beta\,.\,x-\gamma}+\frac{\gamma-a}{x-\gamma\,.\,x-a}+\frac{a-\beta}{x-a\,.\,x-\beta}.$$

12. A quadric function such that the degree of the numerator is less by 4 than that of the denominator is said to be "curtate."

The conditions, in order that the function

$$\left(a,\ b,\ c,\ f,\ g,\ h\middle|\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma}\right)^2$$

may be curtate, are easily found to be

$$a+b+c+2f+2g+2h=0,$$

$$a(a+h+g)+\beta(h+b+f)+\gamma(g+f+c)=0;$$

and by reason of the superfluous constant we are at liberty to assume a third condition: the three conditions may be taken to be $a+h+g$, $h+b+f$, $g+f+c$ each $=0$; and this being so the values of f, g, h are $=\frac{1}{2}(a-b-c)$, $\frac{1}{2}(-a+b-c)$, $\frac{1}{2}(-a-b+c)$ respectively. Hence the form is

$$\left(a,\ b,\ c,\ \tfrac{1}{2}(a-b-c),\ \tfrac{1}{2}(-a+b-c),\ \tfrac{1}{2}(-a-b+c)\middle|\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma}\right)^2,$$

which, as already mentioned, we denote by

$$\left(a,\ b,\ c\,\bar{\cdot}\middle|\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma}\right)^2.$$

We have thus the theorem that a curtate function of any number of inverts, but with only the three essential inverts

$$\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma},$$

is always expressible in the foregoing form

$$\left(a,\ b,\ c\,\bar{\cdot}\middle|\frac{1}{x-a},\ \frac{1}{x-\beta},\ \frac{1}{x-\gamma}\right)^2.$$

13. It may be remarked that the function $(a,\ b,\ c\,\bar{\cdot}\ \mid X,\ Y,\ Z)^2$ is a function of the differences of the variables X, Y, Z; and similarly, in the case of four variables, a function $(a,\ b,\ c,\ d,\ f,\ g,\ h,\ l,\ m,\ n\ \mid X,\ Y,\ Z,\ W)^2$, for which

$$a+h+g+l,\quad h+b+f+m,\quad g+f+c+n,\quad l+m+n+d,$$

are each $= 0$, is a function of the differences of the variables X, Y, Z, W: and so in general. Any such function is said to be "diaphoric": and it is easy to see that, taking for the variables any inverts whatever, a diaphoric function is always curtate.

14. The function

$$\left(\cdots a\cdots, \cdots b\cdots, \cdots c\cdots \big\lvert\!\!\big\lvert \frac{1}{x-a}, \frac{1}{x-a_1}, \cdots\right)^2,$$

where the coefficients $a, b, c, \ldots$ satisfy the relation $a+b+c+\cdots = -2$, is diaphoric, and therefore curtate. In fact, forming the sum, coeff. $\dfrac{1}{(x-a)^2} + \frac{1}{2}$ coeff. $\dfrac{1}{x-a\,.\,x-\beta} + \cdots$, this is $= a + \frac{1}{2}a^2 - \frac{1}{2}ab - \frac{1}{2}ac - \cdots$, $= -\frac{1}{2}a(2+a+b+c+\cdots)$, which is $= 0$; and similarly the other conditions are satisfied.

15. The function

$$\left(\cdots a\cdots, \cdots b\cdots, \cdots c\cdots \big\lvert\!\!\big\lvert \frac{1}{x-a}, \frac{1}{x-a_1}, \cdots, \frac{1}{x-\beta}, \cdots\right)^2$$

regarded as a function of the inverts $\dfrac{1}{x-a}, \dfrac{1}{x-a_1}, \cdots, \dfrac{1}{x-\beta}, \cdots$, where

$$a + a_1 + \cdots = b + b_1 + \cdots = c + c_1 + \cdots, = h \text{ suppose},$$

is diaphoric, and therefore curtate. In fact, the condition in regard to $\dfrac{1}{x-a}$ is

$$a(a+a_1+\cdots) + \tfrac{1}{2}(a-b+c)(ac+ac_1+\cdots) + \tfrac{1}{2}(-a+b-c)(ab+ab_1+\cdots) + \tfrac{1}{2}(-a-b+c)(\cdots) = 0;$$

that is,

$$ah\left\{a + \tfrac{1}{2}(-a+b-c) + \tfrac{1}{2}(-a-b+c)\right\} = 0,$$

which is satisfied. And similarly the other conditions are satisfied.

The functions P, Q, R. Art. Nos. 16 to 20.

16. We consider P, Q, R, rational and integral functions of z, such that $P+Q+R=0$: hence, using the accent to denote differentiation in regard to z, we have also $P'+Q'+R'=0$; and therefore $QR'-Q'R = RP'-R'P = PQ'-P'Q, = \Theta$ suppose: and we require to find P, Q, R, such that the function Θ contains only the factors of P, Q, R.

17. It is to be observed that, effecting upon a solution P, Q, R any linear substitution $\div(\alpha z+\beta)\div(\gamma z+\delta)$, and omitting the common denominator, we have a solution; but this is regarded as identical with the original solution. The three functions, if not originally of the same order, can thus be made to be of the same order; or by taking account of the root $z = \infty$, we may in the original case regard them as being of the same order, and it is convenient so to regard them: say they are taken to be of the same order δ. And there is clearly no loss of generality in taking the three functions to be prime to each other; for any common factor of two of them would divide the third, and might therefore be struck out.

18. We may therefore write

$$P = F\Pi(z-l)^p, \quad Q = G\Pi(z-m)^q, \quad R = H\Pi(z-n)^r,$$

where $(z-l)^p$ is taken to denote the distinct simple or multiple factors of P, and the like as regards Q and R; the factors $z-l$, $z-m$, $z-n$ are thus all of them different. And we have $\delta = \Sigma p, = \Sigma q, = \Sigma r$.

19. It is at once seen that Θ is of the degree $2\delta - 2$, and moreover that it contains the factors $\Pi(z-l)^{p-1}$, $\Pi(z-m)^{q-1}$, $\Pi(z-n)^{r-1}$; hence it contains the factor

$$\Pi(z-l)^{p-1}(z-m)^{q-1}(z-n)^{r-1}.$$

Suppose the number of distinct indices p is $= \sigma_1$, that of distinct indices q is σ_2, and that of distinct indices r is σ_3; then the degree of the factor is $= 3\delta - \sigma_1 - \sigma_2 - \sigma_3$; and if this be $= 2\delta - 2$, then Θ can have no other valuable factor: viz. if the numbers $\sigma_1, \sigma_2, \sigma_3$ of the distinct indices p, q, r respectively are such that $\sigma_1 + \sigma_2 + \sigma_3 = \delta + 2$, a relation which is henceforth taken to be satisfied, then we have

$$\Theta = K\Pi(z-l)^{p-1}(z-m)^{q-1}(z-n)^{r-1}.$$

As already in effect remarked, the conclusion extends to the case where P, Q, R are not of the same degree; the equation $P + Q + R = 0$ here implies that two functions, say P, Q, are of the same degree, and the third function R of an inferior degree; but, this being so, we have only to regard R as containing the factor $\left(1 - \dfrac{z}{\infty}\right)^t$ of the degree t proper for raising its degree up to that of P or Q.

20. Solutions are given in the following PQR-Table; in which, where required, the proper factor $\left(1 - \dfrac{z}{\infty}\right)$ has been added; the first column headed Ref. No. (Reference Number) will be explained further on. The Annex to the same Table will also be explained.

The PQR-Table.

Ref. No.	$P=$	$Q=$	$R=$	$\Theta=$
1	z^n	$-1\left(1-\frac{z}{\infty}\right)^n$	$-z^n+1$	$nz^{n-1}\left(1-\frac{z}{\infty}\right)^{n-1}$
2	$4z^n\left(1-\frac{z}{\infty}\right)^n$	$-(z^n+1)^2$	$(z^n-1)^2$	$-4nz^{n-1}(z^n+2)(z^n-1)\left(1-\frac{z}{\infty}\right)^{n-1}$
$3\frac{1}{2}$	$(z^4+2\sqrt{-3}\,z^2+1)^3$	$-12\sqrt{-3}\,z^2(z^4-1)^2\left(1-\frac{z}{\infty}\right)^2$	$-(z^4-2\sqrt{-3}\,z^2+1)^3$	$24\sqrt{-3}\,(z^4+2\sqrt{-3}\,z^2+1)^2$ $(z^4-2\sqrt{-3}\,z^2+1)^2\,z(z^4-1)\left(1-\frac{z}{\infty}\right)$
4	$(z^8+14z^4+1)^3$	$-(z^{12}-33z^8-33z^4+1)^2$	$-108\,(z^5-z)^4\left(1-\frac{z}{\infty}\right)^4$	$216\,(z^8+\cdots)^2(z^{12}-\cdots)\,(z^5-z)^3\left(1-\frac{z}{\infty}\right)^3$
5	$(z^{20}-228z^{15}+494z^{10}+228z^5+1)^3$	$-(z^{30}-522z^{25}-10005z^{20}+0z^{15}-10005z^{10}+522z^5-1)^2$	$-1728\,(z^{11}+11z^6-z)^5\left(1-\frac{z}{\infty}\right)^5$	$-8640\,(z^{20}-\cdots)^2(z^{11}+\cdots)^4\left(1-\frac{z}{\infty}\right)^4$
III, V, VII, VIII	$4z\left(1-\frac{z}{\infty}\right)$	$-(z+1)^2$	$(z-1)^2$	$-4(z+1)(z-1)$
IX	$(z-4)^3$	$-(z-1)(z+8)^2$	$27\,z^2\left(1-\frac{z}{\infty}\right)$	$27\,(z-4)^2(z+8)\,z$
X	$z(z+8)^3$	$-(z^2-20z-8)^2$	$-64\,(z-1)^3\left(1-\frac{z}{\infty}\right)$	$-64\,(z+8)^2(z^2-20z\cdots)(z-1)$
XI	$4(z^2-z+1)^3$	$-(2z^3-3z^2-3z+2)^2$	$-27\,z^2(z-1)^2\left(1-\frac{z}{\infty}\right)^2$	$-108\,(z^2-\cdots)^2(2z^3-3z^2-\cdots)$ $\cdot\,z(z-1)\left(1-\frac{z}{\infty}\right)$
XII	$z^3(z+5)^2(z+8)$	$-(z^3+9z^2+12z-8)^2$	$-64\,(3z-1)\left(1-\frac{z}{\infty}\right)^5$	$-960\,z^2(z+5)(z^3+9z^2\cdots)\left(1-\frac{z}{\infty}\right)^4$
XIII	$(z^2+14z+1)^3$	$-(z^3-33z+1)^2$	$-108\,z(z-1)^4$	$-108\,(z^2+14z\cdots)^2(z^3-33z\cdots)(z-1)^3$
XIV	$(64z+189)(64z^2+133z+49)^3$	$-z\,(4096z^3+18816z^2+25725z+\cdots)^2$	$-27\cdot 7^7\,(z+1)^3\left(1-\frac{z}{\infty}\right)^5$	$-135\cdot 7^7\,(64z^2+\cdots)^2(4096z^3+\cdots)$ $\cdot\,(z+1)\left(1-\frac{z}{\infty}\right)^4$
XV	$-(5z-27)(125z^3-25z^2-265z-243)^3$	$-(-3125z^5+9375z^4+18750z^3+8750z^2+30750z+\cdots)^2$	$+1\,382\,400\,000\,z^3(z+1)^2\left(1-\frac{z}{\infty}\right)^5$	$-1\,382\,400\,000\,(125z^3-\cdots)^2(-3125z^5+\cdots)z^2(z+1)\left(1-\frac{z}{\infty}\right)^4$

In the second half of the table the functions P, Q, R, except in the lines XII, XIV and XV, were calculated by Brioschi; those for the lines XII and XIV were calculated by Klein, but as regards line XV there would seem to have been some error at the beginning of the calculation, and the values found by him are erroneous.

Annex to the PQR-Table.

Ref. No.	$(a, b, c)^*$	Calculation of (ap), (bq), (cr).			Polygon
1	$\frac{1}{2}(1-n^{-2}),\ \frac{1}{2}(1-n^{-2}),\ 0$	$an = 0$	$bn = 0$	$c1 = 0$	I Polygon
2	$\frac{1}{2}(1-n^{-2}),\ \frac{3}{8},\ \frac{3}{8}$	$an = 0$	$b2 = 0$	$c2 = 0$	I Double Pyramid
3	$\frac{4}{9},\ \frac{3}{8},\ \frac{4}{9}$	$a3 = 0$	$b2 = 0$	$c3 = 0$	II Tetrahedron
4	$\frac{4}{9},\ \frac{3}{8},\ \frac{15}{32}$	$a3 = 0$	$b2 = 0$	$c4 = 0$	IV Cube and Octahedron
5	$\frac{4}{9},\ \frac{3}{8},\ \frac{12}{25}$	$a3 = 0$	$b2 = 0$	$c5 = 0$	VI Dodecahedron and Icosahedron
III	$\frac{4}{9},\ \frac{3}{8},\ \frac{4}{9}$	$a1 = \frac{4}{9},\ a1 = \frac{4}{9}$	$b2 = 0$	$c2 = \frac{5}{18} : \Sigma\left(\frac{1}{z},\ \frac{1}{z-\infty},\ \frac{1}{z-1}\right)^2$	Tetrahedron
V	$\frac{15}{32},\ \frac{3}{8},\ \frac{4}{9}$	$a1 = \frac{15}{32},\ a1 = \frac{15}{32}$	$b2 = 0$	$c2 = \frac{5}{18}$: &c.	Cube and Octahedron
VII	$\frac{4}{9},\ \frac{3}{8},\ \frac{12}{25}$	$a1 = \frac{4}{9},\ a1 = \frac{4}{9}$	$b2 = 0$	$c2 = \frac{21}{50}$	Dodecahedron and Icosahedron
VIII	$\frac{12}{25},\ \frac{3}{8},\ \frac{4}{9}$	$a1 = \frac{12}{25},\ a1 = \frac{12}{25}$	$b2 = 0$	$c2 = \frac{5}{18}$	„
IX	$\frac{4}{9},\ \frac{3}{8},\ \frac{12}{25}$	$a3 = 0$	$b1 = \frac{3}{8},\ b2 = 0$	$c1 = \frac{12}{25},\ c2 = \frac{21}{50}$	„
X	„ „ „	$a1 = \frac{4}{9},\ a3 = 0$	$b2 = 0$	$c1 = \frac{12}{25},\ c3 = \frac{21}{50}$	„
XI	„ „ „	$a3 = 0$	$b2 = 0$	$c2 = \frac{21}{50},\ c2 = \frac{21}{50}$	„
XII	„ „ „	$a1 = \frac{4}{9},\ a2 = \frac{5}{18},\ a3 = 0$	$b2 = 0$	$c1 = \frac{12}{25},\ c5 = 0$	„
XIII	„ „ „	$a3 = 0$	$b2 = 0$	$c1 = \frac{12}{25},\ c1 = \frac{12}{25}$	„
XIV	„ „ „	$a1 = \frac{4}{9},\ a3 = 0$	$b1 = \frac{3}{8},\ b2 = 0$	$c2 = \frac{21}{50},\ c5 = 0$	„
XV	„ „ „	$a1 = \frac{4}{9},\ a3 = 0$	$b2 = 0$	$c2 = \frac{21}{50},\ c3 = \frac{21}{50}$	„

* Observe, as regards the (a, b, c) column, that the line III agrees with 3; line V with 4 (only there is a transposition of $\frac{4}{9}$ and $\frac{15}{32}$); lines VII and VIII agree each of them with 5 (only as regards VIII there is a transposition of $\frac{4}{9}$ and $\frac{12}{25}$): the remaining lines IX to XV agree each of them with 5. Observe also as regards the Annex generally, the Roman numbering of the lines 2, 3, 4, 5; the lines after the first have thus the Roman numbers I to XV, corresponding to the Roman numbers used by Schwarz.

The Differential Equations $\{x,\, z\}$ and $\{s,\, x\}$. Art. Nos. 21 to 45.

21. In reference to what follows, it is convenient to put $P = XP_0$, $P' = X_1P_1$, $= P''$ multiplied by the product $\Pi(z-l)$ of the several factors taken each with the index unity; and so for Q and R: viz. we write

$$P,\, Q,\, R = XP_0,\, YQ_0,\, ZR_0,$$

$$P',\, Q',\, R' = X_1P_0,\, Y_1Q_0,\, Z_1R_0,$$

and the foregoing value of Θ then is

$$\Theta = KP_0Q_0R_0.$$

We come now to the investigation of the leading theorem. Take a, b, c arbitrary, f, g, $h = b{-}c$, $c{-}a$, $a{-}b$; P, Q, R functions of z as above; and write

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R,$$

equations, which are consistent with each other and determine x as a rational function of z. Using, as before, the accent to denote differentiation in regard to z, and taking the coefficients (a, b, c) arbitrary, it is required to find the value of

$$\{x,\, z\} + x'^2\left(a, b, c \,\bar{\therefore}\, \middle|\, \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2.$$

22. Calculation of the first term $\{x,\, z\}$.

We have x a function $(\alpha P + \beta R) \div (\gamma P + \delta R)$, $= \dfrac{P}{R}$ suppose; and thence $\{x,\, z\} = \left\{\dfrac{P}{R},\, z\right\}$; for a moment; then

$$\frac{x'}{x} = \frac{(P/R)'}{P/R} = \frac{PR' - P'R}{PR} = -\frac{P_0Q_0 \cdot KR_0 \cdot}{Z\,Z_1\,R_0\, \cdot\, P/R} = -\frac{P_1Q_0}{Z_1\,R_0}.$$

Substituting the values

$$P_0 = \Pi(z-l)^{p-1}, \quad Q_0 = \Pi(z-m)^{q-1}, \quad R_0 = \Pi(z-n)^{r-1}, \quad Z = \Pi(z-n),$$

we have

$$\frac{x'}{x} = \Sigma\,\frac{p-1}{z-l} + \Sigma\,\frac{q-1}{z-m} - \Sigma\,\frac{r+1}{z-n},$$

and thence

$$\{x,\, z\} = \Sigma\,\frac{p-1}{(z-l)^2} + \Sigma\,\frac{q-1}{(z-m)^2} - \Sigma\,\frac{r+1}{(z-n)^2}$$
$$-\tfrac{1}{2}\left[\Sigma\,\frac{p-1}{z-l} + \Sigma\,\frac{q-1}{z-m} - \Sigma\,\frac{r+1}{z-n}\right]^2,$$

or say

$$\{x,\, z\} = \tfrac{1}{2}\Bigg(\Sigma\,\frac{p_1-1}{(z-l)^2} + \Sigma\,\frac{p_1-1}{(z-l_1)^2} - \Sigma\,\frac{q-1}{(z-m)^2} - \Sigma\,\frac{q-1}{(z-m_1)^2} + \Sigma\,\frac{r+1}{(z-n)^2} + \Sigma\,\frac{r+1}{(z-n_1)^2}$$
$$- \frac{p_1-1}{z-l}\,\frac{p_1-1}{z-l_1} - \cdots - \frac{q_1-1}{z-m}\,\frac{q_1-1}{z-m_1} - \cdots + \frac{r+1}{z-n}\,\frac{r+1}{z-n_1} + \cdots\Bigg).$$

where it is to be observed that

$$\Sigma(p-1)+\Sigma(q-1)-\Sigma(r+1)=\delta-\sigma_1+\delta-\sigma_2-(\delta+\sigma_3)=\delta-\sigma_1-\sigma_2-\sigma_3=-2;$$

consequently the function is diaphoric, and therefore curtate.

It is to be remarked that the function, although presenting itself in a form unsymmetric in regard to the factors of P and Q, and of R, is really symmetric as regards the three sets of factors; this is obvious a priori, and it will be presently verified.

23. For the calculation of the second term

$$x'^2\left(a,b,c\,\bar{\therefore}\,\middle\|\,\frac{1}{x-a},\ \frac{1}{x-b},\ \frac{1}{x-c}\right)^2$$

we have

$$f(x-a),\ g(x-b),\ h(x-c)=\Omega P,\ \Omega Q,\ \Omega R,$$

where Ω is a determinate function of z; hence

$$\frac{x'}{x-a},\ \frac{x'}{x-b},\ \frac{x'}{x-c}=\frac{P'}{P}+\frac{\Omega'}{\Omega},\ \frac{Q'}{Q}+\frac{\Omega'}{\Omega},\ \frac{R'}{R}+\frac{\Omega'}{\Omega}.$$

Then substituting these values, by reason that the function is diaphoric, the terms in $\dfrac{\Omega'}{\Omega}$ disappear, and we have

$$x'^2\left(a,b,c\,\bar{\therefore}\,\middle\|\,\frac{1}{x-a},\ \frac{1}{x-b},\ \frac{1}{x-c}\right)^2=\left(a,b,c\,\bar{\therefore}\,\middle\|\,\frac{P'}{P},\ \frac{Q'}{Q},\ \frac{R'}{R}\right)^2,$$

which is

$$=\left(a,b,c\,\bar{\therefore}\,\middle\|\,\Sigma\,\frac{p}{z-l},\ \Sigma\,\frac{q}{z-m},\ \Sigma\,\frac{r}{z-n}\right)^2.$$

We have $\Sigma p=\Sigma q=\Sigma r,=\delta$: and hence by what precedes, this function, considered as a function of the inverts $\dfrac{1}{z-l}$, &c., is diaphoric, and therefore curtate.

24. We have therefore

$$\{x,\,z\}+x'^2\left(a,b,c\,\bar{\therefore}\,\middle\|\,\frac{1}{x-a},\ \frac{1}{x-b},\ \frac{1}{x-c}\right)^2$$

$$=\tfrac{1}{2}\left[\Sigma\,\frac{p-1}{(z-l)^2}+\Sigma\,\frac{q-1}{(z-m)^2}-\Sigma\,\frac{r+1}{(z-n)^2}\right]$$

$$-\tfrac{1}{2}\left[\Sigma\,\frac{p-1}{z-l}+\Sigma\,\frac{q-1}{z-m}-\Sigma\,\frac{r+1}{z-n}\right]^2+\left(a,b,c\,\bar{\therefore}\,\middle\|\,\Sigma\,\frac{p}{z-l},\ \Sigma\,\frac{q}{z-m},\ \Sigma\,\frac{r}{z-n}\right)^2,$$

where the whole function on the right-hand side is curtate.

25. We have to bring the function on the right-hand side into the reduced form

$$\frac{a}{(z-a)^2}+\cdots+\frac{A}{z-a}+\cdots,$$

for the purpose of getting rid of the non-essential inverts (if any).

We write

$$\begin{aligned}\Sigma\,\frac{p_1-1}{(z-l)^2}&=\frac{p-1}{(z-l)^2}+\frac{p_1-1}{(z-l_1)^2}+\cdots\\&=\frac{p-1}{(z-l)^2}+\Sigma_1\,\frac{p_1-1}{(z-l_1)^2};\end{aligned}$$

viz. $z-l$ here denotes any particular factor, and $z-l_1$ represents any other factor of the same set; and so in other like cases.

26. The whole coefficient of $\dfrac{1}{(z-l)^2}$ is

$$=-(p-1)-\tfrac{1}{2}(p-1)^2+ap^2,\quad =\tfrac{1}{2}(1-p^2)+ap^2;$$

an expression which, regarded as a function of a and p, is represented by (ap): the parentheses are used only to avoid ambiguity, and are omitted when p is a number, thus $a1=a$, $a2=-\frac{3}{2}+4a$, and so in other cases.

27. The whole term in $\dfrac{1}{z-l}$ comes from

$$-\frac{p-1}{z-l}\left[\Sigma_1\,\frac{p_1-1}{z-l_1}+\Sigma\,\frac{q-1}{z-m}-\Sigma\,\frac{r+1}{z-n}\right]$$

$$+\frac{2ap}{z-l}\left[\Sigma_1\,\frac{p}{z-l_1}+\tfrac{1}{2}(-a-b+c)\Sigma\,\frac{q}{z-m}+\tfrac{1}{2}(-a+b-c)\Sigma\,\frac{r}{z-n}\right].$$

Viz. each term such as $-\dfrac{p_1-1}{z-l_1}\cdot\dfrac{1}{z-l}$ is to be replaced by $\dfrac{1}{z-l\,.\,z-l_1}\left(\dfrac{1}{z-l}-\dfrac{1}{z-l_1}\right)$, giving rise to the term $-\dfrac{p_1-1}{l-l_1}\,\dfrac{1}{z-l}$, or contributing the term $-\dfrac{1}{l-l_1}$ to the coefficient of $\dfrac{1}{z-l}$. The whole coefficient thus is

$$=-(p-1)\left(\Sigma_1\,\frac{p_1-1}{l-l_1}+\Sigma\,\frac{q-1}{l-m}-\Sigma\,\frac{r+1}{l-n}\right)$$

$$+2ap\,\Sigma_1\,\frac{p}{l-l_1}+p(-a-b+c)\,\Sigma\,\frac{q}{l-m}+p(-a+b-c)\,\Sigma\,\frac{r}{l-n}.$$

28. Suppose first that $z-l$ is a multiple factor of P, viz. a factor with an index p greater than 1: then, for $z=l$, we have $Q+R=0$, $Q'+R'=0$, and thence $\dfrac{Q}{R}=\dfrac{Q'}{R'}$. We have therefore $\Sigma\,\dfrac{q}{l-m}=\Sigma\,\dfrac{r}{l-n}$; that is,

$$p(-a-b+c)\,\Sigma\,\frac{q}{l-m}+p(-a+b-c)\,\Sigma\,\frac{r}{l-n}$$

moreover, in the top line, the terms $\Sigma\,\dfrac{q}{l-m}$ and $-\Sigma\,\dfrac{r}{l-n}$ destroy each other. The whole coefficient of $\dfrac{1}{z-l}$, when $z-l$ is a multiple factor of P, thus is

$$= -(p-1)\left(\Sigma_1\,\frac{p_1-1}{l-l_1}+\Sigma\,\frac{1}{l-m}-\Sigma\,\frac{1}{l-n}\right)$$

$$+2ap\,\Sigma\,\frac{p_1}{l-l_1}-ap\,\Sigma\left(\frac{q}{l-m}+\Sigma\,\frac{r}{l-n}\right),$$

a form which is now symmetrical in regard to the inverts $\dfrac{1}{l-m}$ and $\dfrac{1}{l-n}$.

29. The value just obtained must be equal to

$$(1-p^2+2ap)\left(\Sigma\,\frac{p_1-1}{l-l_1}+\tfrac{1}{2}\Sigma\,\frac{r-1}{l-n}-\Sigma\,\frac{1}{l-l_1}\right);$$

viz. comparing the two forms and reducing, they will be identical if only

$$(1-p+2ap)\left\{\Sigma\,\frac{p+p_1}{l-l_1}-\Sigma\,\frac{\frac{1}{2}(1+p)q-p}{l-m}-\Sigma\,\frac{\frac{1}{2}(1+p)r-p}{l-n}\right\}=0,$$

and it can be shown that the function inside the [ ] is in fact $=0$.

30. We have, as before, $\Sigma\,\dfrac{q}{l-m}=\Sigma\,\dfrac{r}{l-n}$; or writing each of these quantities $=\Phi$, the equation to be verified is

$$\Sigma\,\frac{p+p_1}{l-l_1}=(p+1)\,\Phi-p\,\Sigma\,\frac{1}{l-m}-p\,\Sigma\,\frac{1}{l-n}.$$

We have

$$\frac{P'}{P}=\frac{P}{z-l}+\Sigma\,\frac{P_1}{z-l_1}=\frac{X_1}{X},$$

that is,

$$\Sigma\,\frac{P_1}{l-l_1}=\left[\frac{X_1}{X}-\frac{p}{z-l}\right]\ \text{for } z=l,$$

$$=\left[\frac{X_1(z-l)-pX}{X(z-l)}\right].$$

The first derived function of the numerator is $X_1'(z-l)+X_1-pX'$, which for $z=l$ is X_1-pX', which is $=0$; and, for the denominator, it is $X'(z-l)+X$, which is also $=0$. Passing to the second derived functions, we find we find in like manner

$$\Sigma \frac{1}{l - l_1} = \tfrac{1}{2}\, \frac{X''}{X'},$$

and we thence obtain (z being always $= l$)

$$\Sigma \frac{p + p_1}{z - l_1} = \frac{X_1'}{X'},$$

so that the equation to be verified becomes

$$\frac{X_1'}{X'} = (p+1)\,\Phi - p\,\Sigma \frac{1}{l - m} - p\,\Sigma \frac{1}{l - n}.$$

31. But from the equation $\Theta = PQ' - P'Q, = KP_0Q_0R_0$, we find $XY_1 - X_1Y = KR_0$, and then, differentiating, $XY_1' + X'Y_1 - X_1'Y - X_1Y' = KR_0'$: writing in these equations $z = l$, they become

$$-X_1Y = KR_0,$$

$$X'Y_1 - X_1'Y = KR_0',$$

so that, dividing the second by the first,

$$-\frac{X'}{X_1}\,\frac{Y_1}{Y} + \frac{X_1'}{X_1} = \frac{R_0'}{R_0};$$

or, recollecting that $X_1 = pX'$ and $\dfrac{Y_1}{Y} = \dfrac{Q'}{Q}$, we have

$$\frac{X_1'}{X'} = p\left(\frac{R_0'}{R_0} - \frac{Y'}{Y}\right) + \frac{Q'}{Q},$$

that is,

$$\frac{X_1'}{X'} = p\left(\Sigma \frac{r-1}{z-n} - \Sigma \frac{1}{z-m}\right) + \Sigma \frac{q}{l-m},$$

$$= (p+1)\,\Phi - p\,\Sigma \frac{1}{l-m} - p\,\Sigma \frac{1}{l-n},$$

the required relation.

32. The result is that, $z - l$ being a multiple factor of P, the coefficient of the term $\dfrac{1}{z-l}$ is

$$= (1 - p^2 + 2ap)\left\{\Sigma \frac{p_1 - 1}{l - l_1} + \tfrac{1}{2}\Sigma \frac{r-1}{l-n} - \Sigma \frac{1}{l - l_1}\right\}$$

$$= 2\,(ap)\left[\tfrac{1}{2}\left(\frac{Q'}{Q} + \frac{R'}{R}\right) - \frac{Y'}{Y} - \frac{Z'}{Z} + \frac{X''}{X'}\right].$$

33. In the case where $z - l$ is a simple factor of P we have $p = 1$, and the coefficient is

$$= \Sigma\Sigma' \frac{p_1}{l - l_1} + (-a - b + c)\,\Sigma \frac{q}{l-m} + (-a + b - c)\,\Sigma \frac{r}{l-n}$$

$$= a\left(2\Sigma' \frac{p_1}{l - l_1} - \Sigma \frac{q}{l-m} - \Sigma \frac{r}{l-n}\right) - (b - c)\left(\Sigma \frac{q}{l-m} - \Sigma \frac{r}{l-n}\right).$$

34. Of course the formulae for the coefficients of $\dfrac{1}{(z-l)^2}$ and $\dfrac{1}{z-l}$ give at once, by a mere change of letters, those for the coefficients of $\dfrac{1}{(z-m)^2}$, $\dfrac{1}{z-m}$, and $\dfrac{1}{(z-n)^2}$, $\dfrac{1}{z-n}$; and the function in question,

$$\{x,\, z\} + x'^2\left(a, b, c \,\bar{\therefore}\Bigg\|\, \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2,$$

is now obtained in the required form

$$= \frac{(ap)}{(z-l)^2} + \cdots + \frac{(bq)}{(z-m)^2} + \cdots + \frac{(cr)}{(z-n)^2} + \cdots + \frac{A}{z-l} + \cdots + \frac{B}{z-m} + \cdots + \frac{C}{z-n} + \cdots,$$

where (ap) denotes $\frac{1}{2}(1-p^2) + ap^2$, and the like for (bq) and (cr); and where, $z-l$ being a multiple factor of P, the coefficient A contains the factor (ap), and similarly for B and C.

35. Suppose that the coefficients a, b, c are no one of them $= 0$; we have $a1, = a$, which does not vanish; that is, $z-l$ being a simple factor of P, the expression contains $\dfrac{1}{(z-l)^2}$, or the invert $\dfrac{1}{z-l}$ is essential: and similarly, $z-m$ being a simple factor of Q, or $z-n$ a simple factor of R, the inverts $\dfrac{1}{z-m}$ and $\dfrac{1}{z-n}$ are essential. But for $z-l$ a multiple factor of P, the coefficient (ap) of the term $\dfrac{1}{(z-l)^2}$ may vanish, viz. this will be the case if $a = \frac{1}{2}\left(1 - \dfrac{1}{p^2}\right)$; and, when this is so, the coefficient A of the corresponding term $\dfrac{1}{z-l}$ also vanishes; that is, $\dfrac{1}{z-l}$ is a non-essential invert. And similarly for any multiple factor $z-m$ of Q or $z-n$ of R, the invert $\dfrac{1}{z-m}$ or $\dfrac{1}{z-n}$ may be non-essential.

36. If P, Q, R contain each of them only multiple factors of the same index, say of the indices p, q, r for the three functions respectively, viz. if the functions are $P(\Pi(z-l))^p$, $G(\Pi(z-m))^q$, $H(\Pi(z-n))^r$, the result contains only the six terms written for a, b, c any given values of them $= \frac{1}{2}\left(1-\dfrac{1}{p^2}\right)$, $\frac{1}{2}\left(1-\dfrac{1}{q^2}\right)$, $\frac{1}{2}\left(1-\dfrac{1}{r^2}\right)$ respectively the result is $= 0$: viz. we then have

$$\{x,\, z\} + x'^2\left(a, b, c \,\bar{\therefore}\Bigg\|\, \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = 0,$$

or we in fact have, for the values of a, b, c in question,

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R$$

of this differential equation of the third order.

37. The reasoning applies directly to lines 2, 3, 4, 5 of the PQR-Table; and with a slight variation to line 1; viz. here the factors of R $(= -z^n + 1)$ are all simple factors, but in virtue of $c = 0$ and $n = b$, the corresponding inverts disappear, and, the other inverts also disappearing, the value of the function in $\{x, z\}$ becomes $\equiv 0$.

Thus line 3 shows that the function x, determined by

$$f(x-a) : g(x-b) : h(x-c) = (z^4 + 2\sqrt{-3}\,z^2 + 1)^3 : -12\sqrt{-3}\,(z^4 - 1)^2 : -(z^4 - 2\sqrt{-3}\,z^2 + 1)^3,$$

satisfies

$$\{x,\, z\} + x'^2\left(\tfrac{4}{9},\, \tfrac{3}{8},\, \tfrac{4}{9} \mathbin{\dot{\bar{.}}} \middle\| \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = 0,$$

and so for any other of the five lines.

38. The indices of the factors of P, Q, R may be such that, for proper values of the coefficients a, b, c, there are in all only three essential inverts, say $\dfrac{1}{x-a_1}$, $\dfrac{1}{x-b_1}$, $\dfrac{1}{x-c_1}$, belonging to the three functions P, Q, R respectively, or it may be two, or three, of them to the same function. When this is so, the function of those inverts is, by what precedes, a curtate function, and it is consequently a function

$$\left(a_1,\, b_1,\, c_1 \mathbin{\dot{\bar{.}}} \middle\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2,$$

where a_1, b_1, c_1 are the values of the three which do not vanish in the series of expressions (ap), (bq), (cr).

The remaining lines (III, V, VII, VIII) and IX to XV of the PQR-Table give such values of P, Q, R, the values of (a, b, c) and the calculation of the values of (ap), (bq), (cr) is shown by the corresponding lines of the Annex. And we have thus values of x determined by the equations

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R,$$

and giving

$$\{x,\, z\} + x'^2\left(a, b, c \mathbin{\dot{\bar{.}}} \middle\| \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = \left(a_1,\, b_1,\, c_1 \mathbin{\dot{\bar{.}}} \middle\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2.$$

39. For instance, from line IX we have

$$f(x-a) : g(x-b) : h(x-c) = (z-4)^3 : -(z-1)(z+8)^2 : 27z^2\left(1 - \frac{z}{\infty}\right),$$

the values of (a, b, c) are $\frac{4}{9}$, $\frac{3}{8}$, $\frac{12}{25}$; and since P, Q, R contain factors with the exponents 3; 1, 2; and 1, 2 respectively, the coefficients which present themselves on the right-hand side are

$$a3;\;\; b1,\; b2;\;\; c1,\; c2,$$

which are

$$= 0;\quad \frac{3}{8},\ 0;\quad \frac{12}{25},\ \frac{21}{50},\ \text{respectively.}$$

Hence writing $a_1,\, b_1,\, c_1 = \frac{3}{8},\ 0;\ \frac{12}{25},\ \frac{21}{50}$, the corresponding inverts are $\frac{1}{x-1},\ \frac{1}{x-\infty},\ \frac{1}{x}$; and the result is

$$\{x,\, z\} + x'^2\left(\frac{4}{9},\, \frac{3}{8},\, \frac{12}{25}\, \bar{\therefore}\Big\|\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = \left(\frac{3}{8},\, \frac{12}{25},\, \frac{21}{50}\, \bar{\therefore}\Big\|\frac{1}{x-1},\, \frac{1}{x-\infty},\, \frac{1}{x}\right)^2.$$

40. It is hardly necessary to remark that an expression

$$\left(a_1,\, b_1,\, c_1\, \bar{\therefore}\Big\|\frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2$$

in fact denotes

$$\frac{a_1}{(x-a_1)^2} + \frac{b_1}{(x-b_1)^2} + \frac{-a_1-b_1+c_1}{(x-a_1)(x-b_1)} + \cdots$$

The particular form of the x inverts is immaterial; we could by a general linear transformation upon the x make them to be $\frac{1}{x-a_1},\ \frac{1}{x-b_1},\ \frac{1}{x-c_1}$ with the (a_1, b_1, c_1) arbitrary; or we can give to the a_1, b_1, c_1 any particular values we please: there would be a propriety in making the inverts to be in every case (as in the foregoing example) $\frac{1}{x},\ \frac{1}{x-\infty},\ \frac{1}{x-1}$; but the numerical work would be troublesome, and it is not worth while to effect it.

41. The conclusion is that lines (III, V, VII, VIII) and IX to XV of the PQR-Table, give, for determinate values of (a, b, c) and (a_1, b_1, c_1), solutions

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R$$

of the equation

$$\{x,\, z\} + x'^2\left(a, b, c\, \bar{\therefore}\Big\|\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = \left(a_1, b_1, c_1\, \bar{\therefore}\Big\|\frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2,$$

where a, b, c, a_1, b_1, c_1 are or can be made arbitrary, but without any real gain of generality herein. This is the Differential Equation $\{x,\, z\}$.

42. Recurring to the results from the Arabic lines of the PQR-Table, but for convenience writing s instead of x, we have

$$f(s-a) : g(s-b) : h(s-c) = P : Q : R,$$

where P, Q, R are now functions of z, a solution of

$$[s,\, z] + \left(\frac{ds}{dz}\right)^2\left(a, b, c\, \bar{\therefore}\Big\|\frac{1}{s-a},\, \frac{1}{s-b},\, \frac{1}{s-c}\right)^2 = 0.$$

But we have

$$\{s,\, x\} = -\left(\frac{dx}{ds}\right)^2 [s,\, z],$$

and the foregoing is therefore a solution of

$$[s,\, x] = \left(a, b, c \;\bar{\cdot}\, \Big\| \frac{1}{s-a},\, \frac{1}{s-b},\, \frac{1}{s-c}\right)^2,$$

a differential equation of the third order. This is the Differential Equation $\{s,\, x\}$.

43. From the Roman lines, if we assume

$$f(s-a) : g(s-b) : h(s-c) = P : Q : R,$$

where $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{R}$ are functions of z, not the same functions than P, Q, R are of s, since they belong to a different line of the Table: we have, as before,

$$[x,\, z] + \left(\frac{dx}{dz}\right)^2 \left(a, b, c \;\bar{\cdot}\, \Big\| \frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2 = \left(a_1, b_1, c_1 \;\bar{\cdot}\, \Big\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2.$$

44. We may combine any such result with a properly selected result of the preceding system, the two results being such that (a, b, c) have the same values in each of them. (See as to this the foot-note referring to the Annex to the PQR-Table.) The last equation then becomes

$$[x,\, z] + \left(\frac{dx}{dz}\right)^2 [s,\, x] = \left(a_1, b_1, c_1 \;\bar{\cdot}\, \Big\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2,$$

or since

$$[x,\, z] + \left(\frac{dx}{dz}\right)^2 [s,\, x] = [s,\, z],$$

this is

$$[s,\, z] = \left(a_1, b_1, c_1 \;\bar{\cdot}\, \Big\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2,$$

the corresponding relation between x, z being of course obtained by the elimination of s from the two sets of equations

$$f(s-a) : g(s-b) : h(s-c) = P : Q : R \quad \text{and} \quad f(x-a) : g(x-b) : h(x-c) = \mathfrak{P} : \mathfrak{Q} : \mathfrak{R},$$

viz. the required relation is

$$P : Q : R = \mathfrak{P} : \mathfrak{Q} : \mathfrak{R},$$

where P, Q, R are functions of s; $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{R}$ functions of x; and, in virtue of

$$P + Q + R = 0, \quad \mathfrak{P} + \mathfrak{Q} + \mathfrak{R} = 0,$$

the relations between equivalents to a single equation between x and s. And writing finally x in place of z, that is, now considering $\mathfrak{P}$, $\mathfrak{Q}$, $\mathfrak{R}$ as functions of x, we have

$$\mathfrak{P} : \mathfrak{Q} : \mathfrak{R} = P : Q : R$$

as a solution of

$$[s,\, x] = \left(a_1, b_1, c_1 \;\bar{\cdot}\, \Big\| \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2,$$

a differential equation of the third order of the foregoing form, $[s,\, x]$ = given function of x, but with different values of the coefficients (a_1, b_1, c_1) instead of (a, b, c).

45. It thus appears that there are in all 16 sets of values of (a, b, c) for which the equation is solved, viz. the 16 sets of values are shown in the right-hand column of the Annex. For greater clearness I exhibit the integral equations as follows:

	Functions of z.		Functions of x.
1	$f(x-a) : g(x-b) : h(x-c) = P : Q : R$ (1)	=	Polygon
I	"	=	Double Pyramid
II	$4z : -(z+1)^2 : (z-1)^2$ (2)	=	Tetrahedron
III	"	=	"
IV	$f(x-a) : g(x-b) : h(x-c) = (z-1)^2 : (z+1)^2$ (4)	=	Cube and Octahedron
V	"	=	"
VI	$f(x-a) : g(x-b) : h(x-c) = (4z+1)^2 : (z-1)^2$ (4)	=	Dodecahedron and Icosahedron
VII	"	=	"
VIII	$(z-1)^2 : -(z+1)^2 : 4z$ (2)	=	"
IX	$P : Q : R$ (IX)	=	"
X	"	=	"
XI	"	=	"
XII	"	=	"
XIII	"	=	"
XIV	"	=	"
XV	"	=	"

The values of the P, Q, R as functions of z, are taken out of the PQR-Table: only in the lines III, V, VII, VIII, where P, Q, R are given in the right-hand column of the Annex. For greater clearness I exhibit the integral equations as follows:

$$= 4x, \quad -(x+1)^2, \quad (x-1)^2,$$

and where, as regards V and VIII, line is a transposition of P and R; 1 have inverted the actual values of the $\mathfrak{P}$-functions. (See as to this the foot-note referring to the Annex.)

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46. Considering the foregoing equation

$$[s,\, x] = \left(a_1,\, b_1,\, c_1 \,\dot{\bar{.}}\, \middle\|\, \frac{1}{x-a_1},\, \frac{1}{x-b_1},\, \frac{1}{x-c_1}\right)^2$$

as a particular case of the equation $[s,\, x] =$ Rational function of x, $= R(x)$ suppose, then we have in 1, I, II, IV, VI solutions of the form $x =$ Rational function of s.

Consider, in general, a solution of this form, $x = F(s)$ a rational function of s: then s is an irrational function of x, and if a, b are any values of s, then $[a_1,\, b_1] = R(x)$, $[a_2,\, b_2] = R(x)$; that is, $[a_2,\, b_2] = [a_1,\, b_1]$, and therefore (ante, No. 7) $a_2 = \dfrac{aa_1 + b}{ca_1 + d}$. And then $x = F(a_2) = F\left(\dfrac{aa_1 + b}{ca_1 + d}\right)$, $= F(a_1)$: viz. $F(s)$ is a rational function of s, transformable into itself by the several homographic transformations of a group of such transformations: viz. taking s a group; and conversely to each homographic transformation of s we have a rational function of s, transformable into itself by the several homographic transformations of a group of such transformations; viz. taking s as the parameter of a point, the points which are between any two roots whatever of the equation $x = F(s)$ there exists a homographic relation of the form in question. Further, if s_1 is a root, the others are $\dfrac{as_1 + b}{cs_1 + d}$ for the several homographic transformations $\dfrac{as + b}{cs + d}$ of the group; and transformable into itself by the several homographic transformations of a group of such transformations occur in the form $x = F(s)$; conversely to each homographic transformation $\dfrac{as + b}{cs + d}$ of the group correspond to the six elements of the rational function $x = F(s)$ a Group; or, what is the same thing, to each homographic transformation $\dfrac{as + b}{cs + d}$ of the group correspond to a Group; or, what is the same thing, to each homographic transformation $\dfrac{as + b}{cs + d}$ of the group correspond to a Group; the rational function $x = F(s)$ is the function corresponding to the Group.

47. We may, in any equation between x and s, consider s as imaginary variables $p + qi$ and $u + vi$ respectively; considering then (p, q) and (u, v) as rect- angular coordinates of points in different planes, we have a first plane the locus of the points s, and a second plane the locus of the points x: there is between the two planes a correspondence which is in fact the correspondence of conformable figures; to the infinitesimal element ds drawn from a point s of the first figure corresponds an infinitesimal element dx drawn from the corresponding point x of the second figure, these elements being in general connected by an equation of the form $dx = \alpha\, ds$ involving s but not ds; or, what is the same thing, the lengths of the ratio of the lengths of the elements being the moduli of α $(a + bi)$, and the inclinations of the directions being the angle of α $(a + bi)$. Passing to the second derived element d_2s of the same thing along ds, d_2x the corresponding consecutive elements of the path of the point s, then the ratio of the lengths of the elements d_2x, d_2s are the corresponding consecutive elements of the path of the point x, the ratio $d_2x : d_2s$ the corresponding consecutive elements of the path of the path of the point s, then the ratio of the lengths of the elements d_2s and d_2x are the corresponding consecutive elements of the path of the point x. In particular, the consecutive elements ds having all the same direction (that is, the same module λ), as λ is a real positive, has $s = a + bi$, b, c are critical points of the kind in question. In fact, writing $x - a = h$, the equation is of the form

$$x - a = iF' \exp i\,(\theta + \theta_1).$$

48. I consider the foregoing equation $[s,\, x] = R(x)$, where $R(x)$ is a rational function, and is now taken to be a real function of x: we may assume $s' = i\rho'\, e^{i\theta}$, where the accents denote differentiation in regard to x, and where ρ', θ', and therefore also ρ, are real functions of x. We have

$$\frac{s''}{s'} = \frac{\rho''}{\rho'} + i\theta'',$$

and thence

$$\frac{s'''}{s'} - \left(\frac{s''}{s'}\right)^2 = \frac{\rho'''}{\rho'} - \left(\frac{\rho''}{\rho'}\right)^2 + \frac{\theta'''}{\theta'} - \left(\frac{\theta''}{\theta'}\right)^2 + i\theta''$$

$$-\tfrac{1}{2}\left(\frac{s''}{s'}\right)^2 = -\tfrac{1}{2}\left(\frac{\rho''}{\rho'}\right)^2 - \tfrac{1}{2}\left(\frac{\theta''}{\theta'}\right)^2 + \tfrac{1}{2}\theta'^2 - i\frac{\rho''}{\rho'}\, i\theta'' - i\frac{\rho''\theta''}{\rho'},$$

and thence

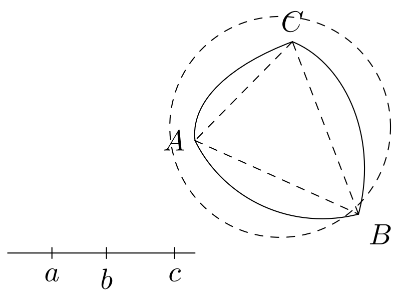
$$[s,\, x] = [\rho,\, x] + [\theta,\, x] + \tfrac{1}{2}\theta'^2 - i\frac{\rho''\theta''}{\rho'\theta'} - i\frac{\rho''\theta''}{\rho'}.$$

Putting this $= R(x)$, and assuming that x is real, we have

$$[\rho,\, x] + [\theta,\, x] + \tfrac{1}{2}\theta'^2 = R(x), \quad 0 = i\frac{\rho''\theta''}{\rho'\theta'}.$$

The last equation gives $\rho''\theta'' = 0$, that is, $\theta'' = 0$, which gives $s' = 0$, and may be disregarded; or else $\rho'' = 0$, therefore ρ', a real constant, $= \gamma$ suppose, and $[\rho,\, x] = 0$; hence for the solution of the equation $[s,\, x] = R(x)$, we have $s' = i\gamma'\, e^{i\theta}$, θ a real quantity determined by $[\theta,\, x] + \frac{1}{2}\theta'^2 = R(x)$; and then, integrating the equation for s', we have $s = a + \beta s + \gamma e^{i\theta}$, a, β, γ real constants.

49. The conclusion is that, if $[s,\, x] = R(x)$, a real function of x, and if x be real, that is, if the point x moves along a right line (say the x-line), then $s = a + \beta s + \gamma e^{i\theta}$ (θ, and the constants a, β, γ being real), that is, the point s moves in a circle, coordinates of the center: a, β, and radius $= \gamma$.



50. Suppose a, b, c are any real values of x representing points a, b, c on the x-line; and A, B, C any given imaginary values of s representing points A, B, C

in the s-plane: since $[s,\, x] = R(x)$ is a differential equation of the third order, the integral contains three arbitrary constants, and we may imagine these so determined that to the values $s = a, b, c$ shall correspond the values $s = A, B, C$ respectively.

If there is not on the x-line any critical point, as the point x moves continuously along this line from the point a towards the point b, the corresponding circle (in- asmuch as a, b, c and A, B, C are corresponding points) must be the circle through the three points A, B, C.

51. If however the points a, b, c are critical points, such that the element ds at the corresponding points A, B, C are equal to multiples of $(dx)^\lambda$, $(dx)^\mu$, $(dx)^\nu$ re- spectively, then on the x-line the point a in the direction ax to a passes continu- ously along the line: the point s at the point A moves continuously along a circle, (in- asmuch as a, b, c and A, B, C are corresponding points) must be the circle through the three points A, B, C.

51. If however the points a, b, c are critical points, such that the elements ds only along the s-line: the point a, b, c are critical points, such that the elements ds along the a-line, $\mu\pi$, $\nu\pi$, of $(dx)^\lambda$, $(dx)^\mu$, $(dx)^\nu$ respectively. The arc ABC becomes continuous ously at a, b, c but the point s at the points A, B, C must change abruptly through the angles $\lambda\pi$, $\mu\pi$, $\nu\pi$, being real), that is, the point s moves in a circle.

51. If however there is on the x-line a critical point, such that the element ds only along the x-line. Pasing the values of λ, μ, ν, being real, and equal to the curvilinear triangle that to the values $x = a, b, c$ shall correspond the values $s = A, B, C$. (In- asmuch as a, b, c and A, B, C are corresponding points) must be the circle through the three points A, B, C.

52. The foregoing equation

$$[s,\, x] = \left(a, b, c \,\bar{\because} \left|\, \frac{1}{x-a}, \frac{1}{x-b}, \frac{1}{x-c}\right)^2\right.$$

where a, b are real values of x representing points a, b, c on the

$$\frac{d}{dx} \log \frac{ds}{ds} = \frac{1-\lambda}{h} + a_1 + a_2 h + a_3 h^2 + \cdots ,$$

which is satisfied. And similarly the other conditions are satisfied.

$$s = hk^{\lambda-1}(1 + b_1 h + b_2 h^2 + \cdots), = k\rho \text{ for shortness.}$$

This is a particular integral, but we have from it the general integral

$$s = \frac{\alpha + \beta k\rho}{\gamma + \delta k\rho}.$$

* Since there is no critical point on the x-line there can be no abrupt change of direction in the path of s, that is, the path of s cannot consist of circular arcs meeting at an angle: but it is in the next further assumed that the path of s cannot consist of different arcs of circles, the one continuing the other without any abrupt change of direction.

If A be the value of s corresponding to a vanishing value of h, then $B = aA$, and we find

$$s = \frac{a + A\delta h\rho}{\gamma + \delta h\rho}, \quad = \left(A + \frac{a}{\delta h\rho}\right)\left(1 + \frac{\gamma}{\delta h\rho}\right)^{-1}, \quad = A + \frac{a - \gamma A}{\delta}\,\frac{1}{h\rho} + \cdots ;$$

viz. reducing $\dfrac{1}{\rho}$ to its principal term h^{λ}, and then writing ds, ds for $s - A$, and h $(= x - a)$ respectively, we have $ds = K\,(dx)^{\lambda}$, or $s - a = \alpha$ is a critical point with the exponent λ; and similarly $x = b$ and $x = c$ are critical points with the exponents μ and ν respectively.

53. Hence in the equation

$$[s,\, x] = \left(a, b, c\,\bar{\because}\middle\|\,\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2,$$

as the point x, passing successively through a, b, c, describes the x-line, the point s, passing successively through A, B, C, describes the circular arcs meeting at the proper triangle ABC. To points s indefinitely near the x-line correspond points x indefinitely near to the boundary ABC of the triangle ABC on the s-plane. Similar reasoning shows that the s-plane, supposing the boundary just spoken of, agrees fully with the s-plane in question, viz. the points A, B, C are critical points of x, with the exponents λ, μ, ν, where a, b, c, λ, μ, ν are connected by the relation

$$f(x-a) : g(x-b) : h(x-c) = P : Q : R$$

of this differential equation of the third order.

54. It is to be remarked, in regard to the values of the values of the values of the values of the values; and conversely, there are two values of x, viz. one in the upper half and the other in the lower half of the x-line, which correspond to the same each of the points s within the curvilinear triangle ABC. The two half-planes (above and below) form, between them, two sets of triangles ABC, &c., and &c., respectively, each of them which are equivalent functions of x. And we have thus the values x, the points A, B, C as the values, on opposite sides of the s-line, of the curvilinear triangles in the upper half-plane corresponds on opposite sides of the x-line, correspond to two sets of triangles ABC, &c., and &c., respectively, certain determinate values; and, in fact, dividing the surface of a sphere into triangles ABC, ABC, &c., alternately of the two kinds: corresponding to all these triangles in the s-plane and the corresponding triangles in the s-plane are these triangles; and hence the shaded half-plane corresponds to one set of them and the unshaded triangles correspond to the other. To the values x, in the upper half of the x-line we have, A, B, C corresponding to the foot-note for the values of $\frac{4}{5}$, $\frac{15}{50}$, etc., and there will be six considerations as these last followers, in the Annex of the PQR-Table; and then showed a *a* priori that the equation

$$[s,\, x] = \left(a, b, c\,\bar{\because}\middle\|\,\frac{1}{x-a},\, \frac{1}{x-b},\, \frac{1}{x-c}\right)^2$$

is algebraically integrable for these values of a, b, c; and only for these values, or for values reducible to them.

55. As an instance, take the double pyramid form: the integral equation is

$$f(x-a) : g(x-b) : h(x-c) = 4x^n : -(x^n-1)^2 : (x^n+1)^2,$$

or say

$$\frac{(x-a)(x-b)}{(x-b)(x-c)} = -\frac{(x^n-1)^2}{(x^n+1)^2};$$

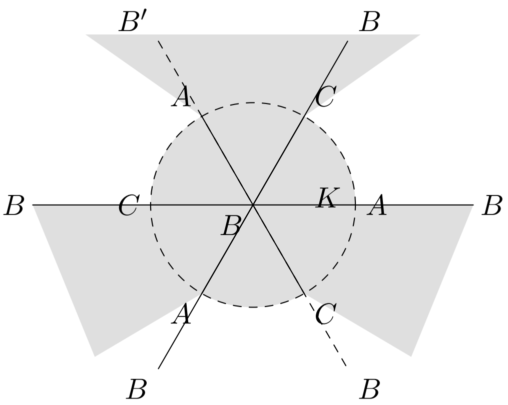
or if, for greater simplicity, we assume a, b, c, $= 1, 0, \infty$, this is $x = \dfrac{(x^n-1)^2}{(x^n+1)^2}$, or say $-(x^n-1) = \sqrt{x}\,(x^n+1)$, that is, $x^n = \dfrac{1 \mp \sqrt{x}}{1 \pm \sqrt{x}}$, a solution of the differential equation

$$[s,\,x] = \left(\frac{3}{8},\ \tfrac{1}{2}(1-n^{-2}),\ \tfrac{3}{8} \therefore \Big|\ \frac{1}{x-1},\ \frac{1}{x-\infty},\ \frac{1}{x}\right)^2.$$

In particular, if $n = 3$, we have $x = \dfrac{(s^3-1)^2}{(s^3+1)^2}$ or $s^3 = \dfrac{1 \mp \sqrt{x}}{1 \pm \sqrt{x}}$, a solution of

$$[s,\,x] = \left(\frac{3}{8},\ \frac{4}{9},\ \frac{3}{8} \therefore \Big|\ \frac{1}{x-1},\ \frac{1}{x-\infty},\ \frac{1}{x}\right)^2.$$

57. We have here the spherical surface divided by the equator and three meridians into twelve triangles, each with the angles $\frac{1}{2}\pi$, $\frac{1}{3}\pi$, $\frac{1}{2}\pi$; and then, projecting from the South pole on the plane of the equator, we have the annexed figure of the s-plane,



divided into 12 curvilinear triangles, each with these same angles $30°, 90°, 60°$; the plane is divided by the shading into two systems, each of 6 triangles. The figure of the s-plane is by the s-line divided into two half-planes, one shaded, the other unshaded; and we have on the line the point c at ∞, a at the origin, and b at the distance unity.

58. Take x real; then, if x is positive and less than 1, s^3 is real and positive, and we have for s the infinitum half-lines at the inclinations 0°, 180°, 300°. If x is real and negative, or in particular greater than 1, viz. here the factors of R $(= -x^n + 1)$ are all simple factors, but in virtue of $c = 0$ and $n = b$, the corresponding inverts disappear, and, the the form $\dfrac{1-hi}{1+hi}$, $= \cos\theta + i\sin\theta$; whence as x is of the same form, or the locus of the point s is a circle radius unity. Writing $s^3 = \dfrac{1-\sqrt{x}}{1+\sqrt{x}}$, and supposing that the point x moves along the x-line through 1 through ∞ to ∞, and thence from ∞ through 0 to b, the point s describes the sides BA, AC, CB of the shaded triangle marked K.

59. Suppose that the point x is at k, in the shaded half-plane and at indefinitely small distance ϵ from the x-line; in the shaded line in question, in the shaded triangle $s\,(-i)$, we have $s^3 = \dfrac{1-\sqrt{x}}{1+\sqrt{x}}$, and then $x = -\frac{1}{4}\epsilon(1-i)$ nearly, and hence a value of s is $s = \dfrac{1-ki}{1+ki}$, $= \cos\theta + i\sin\theta$; whence s^3 is of the same form, or the locus of the point s is a circle radius unity. Writing $s^3 = \dfrac{1-\sqrt{x}}{1+\sqrt{x}}$, and supposing that the point x moves along the x-line a through b through a to ∞, and to ∞ from ∞ to b, the shaded value $\sqrt{x}$ we have $s^3 = \dfrac{1-\epsilon(1-i)}{1+\epsilon(1-i)}$, $= 1 - 2\epsilon(1-i)$ nearly, and hence a value of s is $\epsilon = 1 - \frac{2}{3}\epsilon + \frac{2}{3}\epsilon i$ nearly, and we have to find from a at c in K, and b at the situate on the circle in question.

59. Suppose the point x is at k, in the shaded half-plane, in the shaded triangle small distance ϵ from the x-line, in the shaded line in question, then for the value $x\,(-a) = h$, the shaded line BA, AC, CB of the equation, which $\dfrac{1}{x-a}$ value $\epsilon\,(1-i)$, we have $s^n = \dfrac{1-\epsilon(1-i)}{1+\epsilon(1-i)}$, $= 1 - 2\epsilon(1-i)$ nearly, and hence a value of s is $s = 1 - \frac{2}{3}\epsilon + \frac{2}{3}\epsilon i$, which belongs to the shaded triangle K; and similarly the other triangles; and hence the shaded half-plane corresponds to the shaded triangles, and the unshaded half-plane corresponds to the other.

60. Suppose the point x, instead of being in question of a, b, c, is in the circle in question.

We have here $x - a = a\,(e^{i\theta_1} - e^{-i\theta_1}) = -2ia\,e^{i\frac{1}{2}\pi}\sin\frac{1}{2}\theta_1$; $e^{i\theta_1}\cdot e^{i\theta_2}$, say $x - a = a\,e^{i\theta_1}\cdot e^{i\theta_2}\,e^{i\frac{1}{2}\pi - \frac{1}{2}\theta_2}$ similarly

$$x - a = iP_1\,\exp\tfrac{1}{2}(\theta+\theta_1),$$

Similarly $x - b = iQ_1\,\exp\frac{1}{2}(\theta+\theta_2)$, and $x - c = iR_1\,\exp\frac{1}{2}(\theta+\theta_3)$, where P, Q, R denote $\sin\frac{1}{2}(\theta-\theta_1)$, $\sin\frac{1}{2}(\theta-\theta_2)$, $\sin\frac{1}{2}(\theta-\theta_3)$ respectively. In like manner, $a-b$, $b-c$, $c-a$, $iH\exp\frac{1}{2}(\theta_1+\theta_2)$, $iH\exp\frac{1}{2}(\theta_2+\theta_3)$, $iH\exp\frac{1}{2}(\theta_3+\theta_1)$, where F, G, H denote $\sin\frac{1}{2}(\theta_1-\theta_2)$, $\sin\frac{1}{2}(\theta_2-\theta_3)$, $\sin\frac{1}{2}(\theta_3-\theta_1)$ respectively.

We have

$$-\frac{b-x\,.\,x-a\,.\,a-b}{x-a\,.\,x-b} = \frac{-FGH}{PQR}\exp i\tfrac{1}{2}\,(\theta_1+\theta_2+\theta_3-3\theta),$$

$$\frac{1}{b-c\,.\,x-a} = \frac{-1}{c^2IP}\exp i-\tfrac{1}{2}\,(\theta_2+\theta_3+\theta_1+\theta),$$

with the like values for $\dfrac{1}{c-a\,.\,x-b}$ and $\dfrac{1}{a-b\,.\,x-c}$. Hence the right-hand side of the equation is

$$=\frac{FGH}{PQR}\left(\frac{a}{PP}+\frac{b}{QQ}+\frac{c}{RR}\right)\exp i\,(-2\theta).$$

62. Considering now the left-hand side of the equation, we have

$$[s,\,x] = \frac{1}{\left(\dfrac{dx}{d\theta}\right)^2}\,(\{s,\,\theta\}-\{x,\,\theta\});$$

substituting for x its value $= a+\beta i\,e^{i\theta}+\gamma e^{i\theta}$, this becomes

$$[s,\,x] = -\frac{1}{\gamma^2}\,e^{-i\theta}\,(\{s,\,\theta\}-\tfrac{1}{2}),$$

that is,

$$=-\frac{1}{\gamma^2}\,(\{s,\,\theta\}-\tfrac{1}{2})\exp i\,(-2\theta).$$

Assume $s = L+Mi+Ne^{i\theta}$, L, M and N constants; then using the accent to denote differentiation in regard to θ, we find without difficulty $[s,\,\theta] = (\Theta_1+\Theta_2+\frac{1}{2}\Theta^2)$, and the value of $[s,\,x]$ becomes

$$=-\frac{1}{\gamma^2}\big((\Theta_1+\Theta_2+\tfrac{1}{2}\Theta^2)-\tfrac{1}{2}\big)\exp i\,(-2\theta).$$

Hence, substituting the values of the two sides of the equation, the imaginary factor $\exp i\,(-2\theta)$ divides out, and the equation becomes

$$(\Theta_1+\Theta_2+\tfrac{1}{2}\Theta^2) = \frac{FGH}{PQR}\left(\frac{a}{PP}+\frac{b}{QQ}+\frac{c}{RR}\right),$$

an equation, in which everything is real and which thus determines Θ as a real function of θ; and we have therefore the theorem in question.

Connexion with the differential equation for the hypergeometric series. Art. Nos. 63 to 68.

63. Take p, q given functions of x, and y a function of x determined by the equation

$$\frac{d^2y}{dx^2}+p\,\frac{dy}{dx}+qy=0;$$